

Design and Magnetostatic Simulation of a Solenoid (ANSYS Maxwell 3D)

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Abstract

This project models, simulates and analyses a solenoid in **ANSYS Maxwell 3D** using magnetostatic analysis. The work covers geometry creation (Polygon Helix coil), environment/region definition, conduction path extraction, material assignment, excitation application, meshing strategy, and verification against analytical relations. The report documents the full workflow, provides analytic checks (with worked calculations), suggests parametric studies and next steps (optimization, transient/eddy current analysis, hardware validation).

1. Aim

To design and simulate a solenoid in ANSYS Maxwell 3D; extract magnetic field distribution and basic electromagnetic parameters (B-field, inductance, stored energy); prepare a workflow junior members can reproduce and extend.

2. Objectives

1. Create a helical solenoid geometry (Polygon Helix) with defined dimensions.
2. Define a suitable simulation region (vacuum/air) and boundary.
3. Assign materials (copper, perfect conductor/air) and conduction paths.
4. Apply DC current excitations and run magnetostatic analysis.

5. Extract B and H field plots, compute inductance and stored magnetic energy; validate against analytical formulae.
6. Provide step-by-step instructions, video resources, and a folder structure for team submissions.

3. Background Theory (brief)

- Magnetic field inside an ideal long solenoid:

$$B = \mu_0 n I$$

where $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, n is turns per unit length, I is current.

- Inductance of an (approx.) air-core solenoid:

$$L \approx \frac{\mu_0 N^2 A}{\ell}$$

where N = total turns, A = cross-sectional area, ℓ = solenoid length.

These relations give first-order checks for the simulation.

4. Design Parameters (as used in videos)

- Coil primitive: **Polygon Helix (segmented helix)**
 - Polygon radius = **0.5** (unit)
 - Start radius = **5** (unit)
 - Radius change = **0**
 - Pitch = **1.5** (unit per turn)
 - Turns = **20**
 - Segments = **0**

- Conduction path geometry: **rectangular cross section: 0.5×0.5** (units)
- Region offsets (from transcript): positive X offset 100; other offsets 50 (applied to region)
- Example current excitation used in tutorials: **120 A** (applied to terminals)
- Materials: **Copper** for solenoid; **Perfect Conductor** for terminals/conductor parts; region set as **vacuum/air**

Note: “units” above follow whatever units you use consistently in Maxwell (mm/m)
— be consistent in the model.

5. Step-by-step Modelling Procedure (detailed)

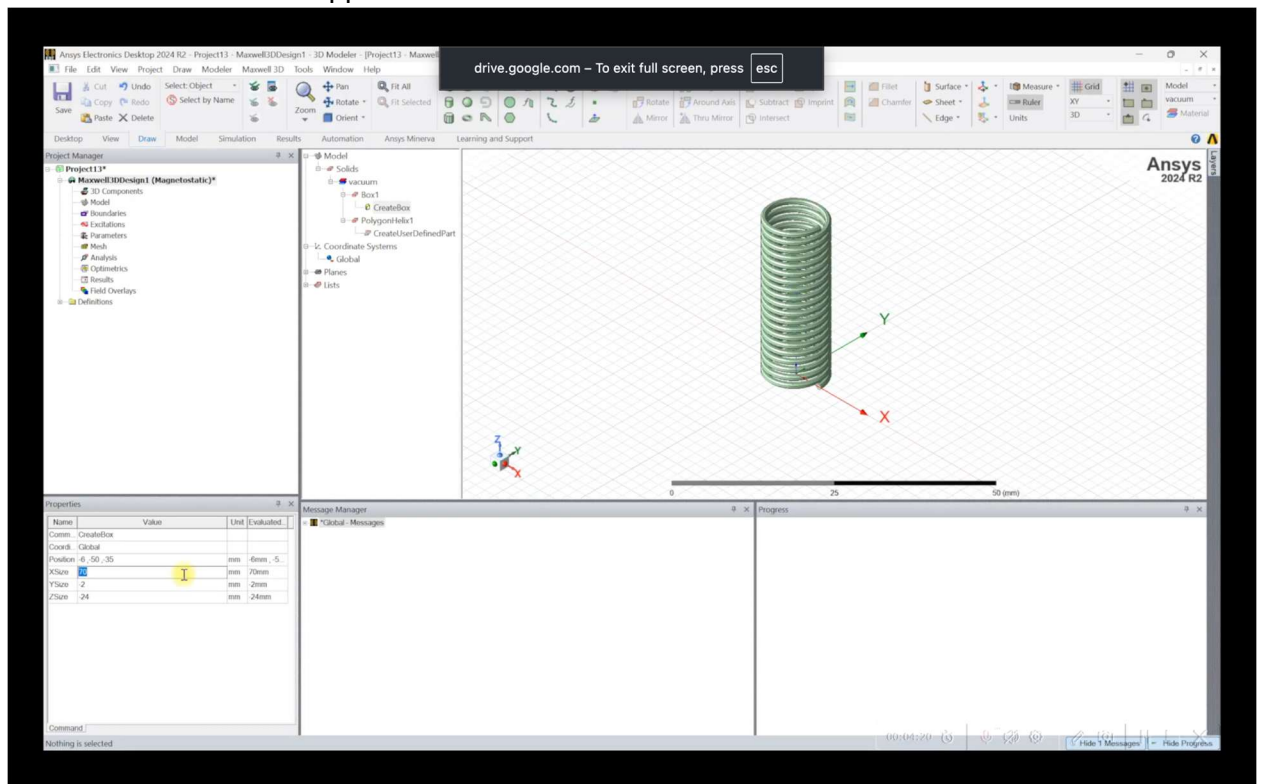
5.1 Project & Interface

1. Open **Ansys Electronics Desktop 2024**.
2. Insert → **Maxwell 3D** design. Leave the default definitions (Model tree, Boundaries, Excitations, Parameters, Mesh, Analysis, Coordinate System) intact.

5.2 Coil geometry (Polygon Helix)

1. Draw → **User Defined Primitive** → **Polygon Helix** (or Segmented Helix).
2. Enter the parameters listed in Section 4 (radius 0.5, start radius 5, pitch 1.5, turns 20, segments 0).

3. Click **OK** → helical coil appears. Rotate/orient view as needed.

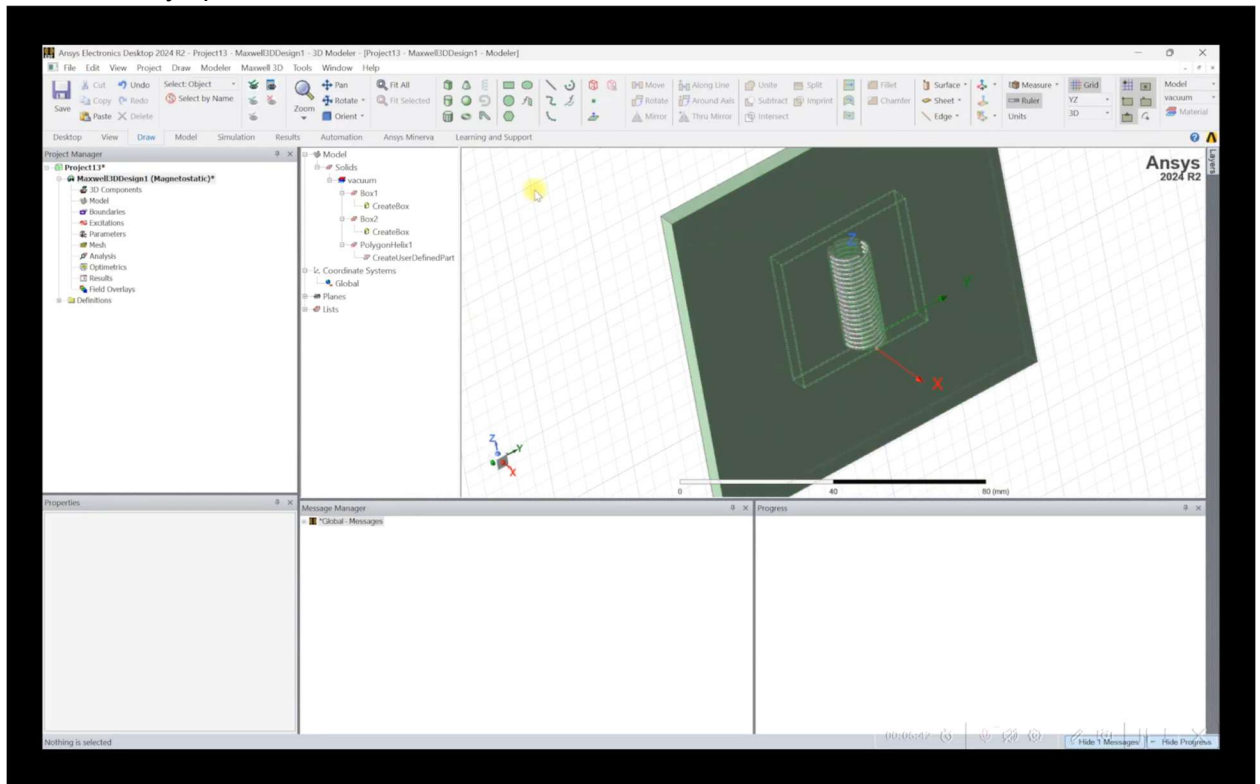


(Fig.1 placeholder: Helical coil view and model tree)

5.3 Surrounding box / region

1. Draw → **Box**. Set position and dimensions to enclose coil with margin. (Transcript used coordinates like position -6, -50, -35 and large extents — adjust for your actual unit system.)

2. Create a smaller box (Box2) and **Subtract** from the large box if needed to create the correct cavity/space.



(Fig.2 placeholder: Box region and coil inside)

5.4 Materials & coloring

1. Select the coil → Properties → rename to **Solenoid** → Material = **Copper** (assign from material library).
2. Select the conductor box → rename **Conductor** → Material = **Perfect Conductor**.
3. Optionally color the coil copper color for clarity.

5.5 Creating conduction paths

1. Select coil face → **Create Polyline** (select center line, mark the conductor path).
2. For each terminal path, set the polyline offset in negative X (transcript used **-15** and then extended to **-5** or **-6** to avoid intersections).

3. Convert 2D polyline to 3D rectangular conductor: cross-section 0.5×0.5 (ensures a volumetric conductor the solver recognizes).
4. Ensure conductor does **not** intersect coil turns (make conductor transparent to visually confirm). If intersecting, extend to outside coil radius until clear.

Important: The order of *Unite/Subtract* matters — when uniting polylines to the solenoid, union order affects material assignment (polylines → solenoid or solenoid → polylines). Follow the correct sequence: select **solenoid first**, then polylines, then **Unite** so polylines adopt copper material (become part of the coil).

5.6 Region/environment

1. Create **Region** → default padding offset 0 to capture all objects.
2. Recreate region with explicit padding: e.g., +X = 100, others = 50 (transcript). This makes a larger vacuum box around the coil.

5.7 Excitations (current terminals)

1. Click the face that will serve as terminal → right click → **Assign excitation** → **Current**.
2. Example: set **Out terminal** = 120 A, **In terminal** = 120 A (or set +120, -120 accordingly).
3. Optionally set current as variable (I) to run parametric sweeps (0 → 120 A in steps).

(Tip: set direction carefully; if directions are swapped, change terminal orientation in the Project Manager → Excitations.)

6. Meshing and Solver Settings

6.1 Meshing

- Use the built-in automated mesh initially.
- Apply **local mesh refinement** near conductors and coil turns: target element size ~ **0.05–0.2 units** (depending on overall geometry size).

- For coarse check: global element size = 0.5; refine to 0.1 for production runs.
- Use a **symmetric** mesh if geometry permits (saves time).

6.2 Solver settings

- Analysis type: **Magnetostatic** (DC steady-state).
- Solver: Adaptive with default convergence; tighten tolerances if results show large residuals.
- For parameter sweeps (current variable), use Parametric or Optimetrics module; save results for each step.

7. Results to Extract & Post-processing

- **Magnetic Flux Density (B)** — vector plot, surface plot inside coil.
- **Magnetic Field Intensity (H)** — vector/contour.
- **Magnetic flux lines / Streamlines** — visualize inside and outside coil.
- **Inductance (L)** — use Maxwell built-in report or compute from coenergy/flux linkage.
- **Stored magnetic energy** — Maxwell provides stored energy; can be computed to $W = \frac{1}{2}LI^2$ cross-check.
- **Parameter sweeps** — B vs I, L vs number of turns, etc.

8. Analytical checks (worked calculations)

Use the design parameters to compute quick analytical checks. All arithmetic shown step-by-step.

8.1 Turns per unit length n

- Pitch $p = 1.5$ units/turn
- Turns $N = 20$
- Length $\ell = p \times N = 1.5 \times 20 = 30$ units
 - Calculation: $1.5 \times 20 = 30$.
- Turns per unit length $n = \frac{N}{\ell} = \frac{20}{30} = 0.6666666667$ turns/unit.
 - Calculation: $20 \div 30 = 0.6666666667$.

8.2 Magnetic flux density inside solenoid (theoretical) for $I = 120$ A

- $\mu_0 = 4\pi \times 10^{-7}$ H/m
 - Numerically: $4 \times \pi = 4 \times 3.141592653589793 = 12.566370614359172$.
 - So $\mu_0 = 12.566370614359172 \times 10^{-7} = 1.2566370614359172 \times 10^{-6}$ H/m.
- Compute $n \times I = 0.6666666667 \times 120 = 80.0$.
 - Calculation: $0.6666666667 \times 120 = 80$.
- Then $B = \mu_0 n I = 1.2566370614359172 \times 10^{-6} \times 80$.
 - Multiply: $1.2566370614359172 \times 10^{-6} \times 80 = 1.0053096491487338 \times 10^{-4}$ T.
- Final numerical result: $B \approx 1.0053 \times 10^{-4}$ T = $100.53 \mu\text{T}$.
 - Interpretation: For $I = 120$ A and given coil geometry, the ideal solenoid formula predicts $\sim 100 \mu\text{T}$ inside; simulation will differ due to finite length, coil spacing, core.

8.3 Approximate inductance L (air-core formula)

- $L \approx \frac{\mu_0 N^2 A}{\ell}$
- Cross-sectional area $A = \pi r^2$ where $r = 5$ units (start radius)
 - $r^2 = 5^2 = 25$.
 - $A = \pi \times 25 = 78.53981633974483$ (since $\pi \approx 3.141592653589793$).
- Compute numerator: $\mu_0 N^2 A$
 - $N^2 = 20^2 = 400$.
 - $400 \times 78.53981633974483 = 31415.92653589793$.
 - Multiply by μ_0 : $31415.92653589793 \times 1.2566370614359172 \times 10^{-6} = 0.03947841760435743$.
- Divide by length $\ell = 30$:
 - $L = 0.03947841760435743 \div 30 = 0.0013159472534785809$ H.
- Final result: $L \approx 1.316 \times 10^{-3}$ H = 1.316 mH.

8.4 Stored magnetic energy at $I = 120$ A

- $W = \frac{1}{2} L I^2$.
- Compute $I^2 = 120^2 = 14400$.
- $\frac{1}{2} L = 0.5 \times 0.0013159472534785809 = 0.0006579736267392904$.
- Multiply: $W = 0.0006579736267392904 \times 14400 = 9.474820225045782$ J.
- Final result: $W \approx 9.475$ J.

These analytical numbers are baseline checks — simulation values will differ due to finite length effects, conductor thickness, region bounding a mesh.

9. Validation Plan

- **Check 1:** Field inside the central region — compare average B (simulation) to analytical B (Section 8.2). Expect simulation B slightly lower (edge effects) for finite length.
- **Check 2:** Inductance from Maxwell (coenergy/flux linkage) vs approximate L computed above — expect simulation L to be within $\pm 20\text{--}30\%$ depending on geometry fidelity and mesh.
- **Check 3:** Energy consistency: compute W from Maxwell and compare to $\frac{1}{2} L^2$ from measured L .
- **Check 4:** Mesh convergence: refine mesh near conductor and re-run; ensure results (B at center, L) converge within tolerance ($\sim 2\text{--}5\%$).

10. Parametric Studies (proposed)

Run sweeps on:

1. **Current (I):** $0 \rightarrow 120$ A in steps of 5 A (as suggested in transcript). Plot B vs I at coil center.
2. **Number of turns (N):** e.g., 10, 20, 30 — check $L \sim N^2$ dependence.
3. **Pitch (p):** tighter vs looser winding: effect on B distribution.
4. **Core material:** Air vs Fe or ferrite (include BH curve for saturation effects).
5. **Conductor cross-section:** effect of conductor width/thickness on proximity/skin (for AC runs later).

11. Common Issues & Troubleshooting

- **Conduction path not recognized:** ensure polylines are converted to volumetric conductor (rectangle cross-section) and united correctly.

- **Wrong material after unite/subtract:** order of operations matters — follow the sequence documented in the video.
- **Excitation direction wrong:** check terminal orientation in project manager; swap direction or reassign if reversed.
- **Mesh/solver errors:** use message manager to read warnings; refine mesh; check geometry overlaps.
- **Large file / students can't install pro version:** use Student Edition; modify simulations to remain within student limits.

12. Practical (Hardware) Extension (only if further extended)

If hardware construction is planned later:

- Wind a physical solenoid with same dimensions ($r = 5$ units, $N = 20$, pitch = 1.5).
- Measure magnetic field with a gaussmeter along the axis and compare to simulation.
- Manufacture a simple iron core to measure enhancement and compare to simulated core insertion.

14. Assessment & Rubric (suggested)

- **Geometry accuracy (30%)** — correct helix params, region, conductor paths.
- **Material & excitation (20%)** — correct materials, terminal assignments and current.
- **Analysis & results (30%)** — B plots, inductance estimate, parametric results.
- **Documentation (20%)** — project files, screenshots, short report, timely uploads.

15. Future Work / Extensions

- **AC transient / Eddy current analysis** (use Eddy Current solver).
 - **Core material study & non-linear BH curves (saturation)**.
 - **Optimization** using parameter sweep or scripting (IronPython) for max B with minimal copper.
 - **Build physical prototype** and validate simulation vs measurement.
 - **Couple with circuit simulation** to design driver for coil (pulse, PWM, etc.).
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16. References & Resources

- ANSYS Maxwell 3D Documentation (ANSYS Electronics Desktop 2024).
- Transcript & tutorial videos (provided): [Solenoid_upto_core](#) and [Solenoid_conduction_path](#).
- Textbooks: J. D. Kraus, *Electromagnetics*; D. J. Griffiths, *Introduction to Electrodynamics*.